

Checkpoint-Based Traverse Verification Without GPS

The Moon does not provide GPS-based positioning. Therefore, after generating the traverse path, the rover must periodically verify that it is still following the planned trajectory during autonomous movement.

Core Idea

The generated path is divided into multiple verification checkpoints. Each checkpoint acts as a planned reference location where the rover confirms its position and orientation.

Checkpoint Verification Method

At each checkpoint, the rover compares expected and observed conditions using:

Verification Input	Purpose
Rover camera feed	Matches local terrain appearance
Sun angle estimation	Verifies orientation and expected lighting geometry
Nearby crater / boulder patterns	Confirms surrounding landmarks
DEM-matched terrain features	Compares expected and observed surface structure
Object detection model	Handles local obstacle avoidance while following path
Expected heading	Confirms rover is moving along correct trajectory

Rover Decision Logic

If checkpoint verification succeeds:

- Rover continues toward the next checkpoint.

If checkpoint verification fails:

- Rover assumes trajectory drift or localization error.
- Current estimated position is sent to the orbiter.
- The orbiter relays the data to Earth.
- A new optimized path is generated from the rover's updated position.
- Updated traverse path is transmitted back to the rover.

Output

A GPS-free path-following strategy that keeps the rover aligned with the planned traverse while allowing recovery through replanning if drift occurs.